Written Assignment 4: Solutions

Q1. K-means Clustering (25 points)

Dataset X **:**
$(4.6, 2.9), (4.7, 3.2), (4.9, 3.1), (5.0, 3.0), (5.1, 3.8), (5.5, 4.2), (5.9, 3.2), (6.0, 3.0), (6.2, 2.8), (6.2, 3.2), (6.5, 3.0), (6.7, 3.1)$

Initial Centers: $\mu_1 = (6.2, 3.2)$ (Red), $\mu_2 = (6.6, 3.7)$ (Green), $\mu_3 = (6.5, 3.0)$ (Blue)

(a) Center of the first cluster (red) after one iteration

Assignment (Iteration 1): Using Euclidean distance $d = \sqrt{\sum (x_i - y_i)^2}$, we assign each point to the nearest center. For the point $(5.1, 3.8)$, the distance to Red is ≈ 1.253 and to Green is ≈ 1.503, making it part of the Red cluster.

Points in Red:
$\{(4.6, 2.9), (4.7, 3.2), (4.9, 3.1), (5.0, 3.0), (5.1, 3.8), (5.9, 3.2), (6.0, 3.0), (6.2, 3.2)\}$

Computation:

- $x = (4.6 + 4.7 + 4.9 + 5.0 + 5.1 + 5.9 + 6.0 + 6.2)/8 = 42.4/8 = 5.300$
- $y = (2.9 + 3.2 + 3.1 + 3.0 + 3.8 + 3.2 + 3.0 + 3.2)/8 = 25.4/8 = 3.175$

Result: $[5.300, 3.175]$

(b) Center of the second cluster (green) after two iterations

Iteration 1 Update: Green initially only captured $(5.5, 4.2)$, so $\mu_2' = [5.5, 4.2]$.

Assignment (Iteration 2): Point $(5.1, 3.8)$ is now closer to Green $(d \approx 0.566)$ than the updated Red center $(d \approx 0.656)$.

Points in Green: $\{(5.1, 3.8), (5.5, 4.2)\}$

Computation:

- $x = (5.1 + 5.5)/2 = 5.300$

- $y = (3.8 + 4.2)/2 = 4.000$

Result: $[5.300, 4.000]$

(c) Center of the third cluster (blue) when the clustering converges

Convergence: The assignments stabilize after 4 iterations. At this stage, the Blue cluster consists of:

$$\{(6.2, 2.8), (6.2, 3.2), (6.5, 3.0), (6.7, 3.1)\}$$

Computation:

- $x = (6.2 + 6.2 + 6.5 + 6.7)/4 = 25.6/4 = 6.400$
- $y = (2.8 + 3.2 + 3.0 + 3.1)/4 = 12.1/4 = 3.025$

Result: $[6.400, 3.025]$

(d) Iterations required for convergence

The cluster assignments stop changing after **4 iterations**.

Q2. Hierarchical Clustering (15 points)

Cluster A (Red): $\{(4.7, 3.2), (4.9, 3.1), (5.0, 3.0), (4.6, 2.9)\}$

Cluster B (Blue): $\{(5.9, 3.2), (6.7, 3.1), (6.0, 3.0), (6.2, 2.8)\}$

(a) Complete Linkage

$d_{complete}(A, B) = \max_{a \in A, b \in B} d(a, b)$.

The maximum distance is between $(4.6, 2.9)$ and $(6.7, 3.1)$:

$$d = \sqrt{(6.7 - 4.6)^2 + (3.1 - 2.9)^2} = \sqrt{2.1^2 + 0.2^2} \approx 2.110$$

Result: 2.110

(b) Single Linkage

$d_{single}(A, B) = \min_{a \in A, b \in B} d(a, b)$.

The minimum distance is between $(5.0, 3.0)$ and $(6.0, 3.0)$:

$$d = \sqrt{(6.0 - 5.0)^2 + (3.0 - 3.0)^2} = 1.000$$

Result: 1.000

(c) Average Linkage

$d_{avg}(A, B) = \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} d(a, b)$.

The mean of all 16 pairwise distances is calculated.

Result: 1.493

Q3. Spectral Clustering (30 points)

(a) Laplacian Matrix

The graph has three connected components: $C_1 = \{1, 2, 3\}$ (triangle), $C_2 = \{4, 5\}$ (edge), and $C_3 = \{6, 7\}$ (edge). The Laplacian $L = D - A$ is:

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 \end{pmatrix}$$

(b) First three eigenvectors

Since there are 3 connected components, the first 3 eigenvalues are $\lambda_1 = \lambda_2 = \lambda_3 = 0$.

The corresponding normalized indicator eigenvectors are:

- $v_1 = [0.577, 0.577, 0.577, 0, 0, 0, 0]^T$
- $v_2 = [0, 0, 0, 0.707, 0.707, 0, 0]^T$
- $v_3 = [0, 0, 0, 0, 0, 0.707, 0.707]^T$

(c) Spectral Embedding

Each node i is represented by the i-th row of the matrix formed by the first three eigenvectors:

- **Nodes 1, 2, 3:** $(0.577, 0, 0)$
- **Nodes 4, 5:** $(0, 0.707, 0)$
- **Nodes 6, 7:** $(0, 0, 0.707)$

Q4. Principal Component Analysis (PCA) (30 points)

Points: $(-1, -1), (0.5, -0.5), (1, 1), (-0.5, 0.5)$. **Mean:** $(0, 0)$.

(a) First Principal Component

The covariance matrix $C = \frac{1}{n} Z^T Z$:

$$C = \begin{pmatrix} 0.625 & 0.375 \\ 0.375 & 0.625 \end{pmatrix}$$

The largest eigenvalue $\lambda_1 = 1.0$ yields the eigenvector:

$$PC_1 = [1/\sqrt{2}, 1/\sqrt{2}]^T \approx [0.707, 0.707]^T$$

Result: $[0.707, 0.707]$

(b) 1-d Representation

Projection onto PC_1:

1. $(-1, -1) \cdot [0.707, 0.707] = -1.414$
2. $(0.5, -0.5) \cdot [0.707, 0.707] = 0.000$
3. $(1, 1) \cdot [0.707, 0.707] = 1.414$
4. $(-0.5, 0.5) \cdot [0.707, 0.707] = 0.000$

New Representations: $\{-1.414, 0.000, 1.414, 0.000\}$